

the Faster Adoption and Manufacturing of (Hybrid &) Electric Vehicles in India (FAME India) has been allocated to set-



ting up approximately 70,000 charging facilities for EVs across the country, this still does not resolve the conundrum of a lack of parking spaces for EVs in office as well as residential areas. Since many four wheeler vehicle owners in India often opt for uncovered parking spots, provided they have access to a dedicated parking space at all, they face issues with regards to the lack of proximity to a charging outlet for their EVs.

CHARGER STANDARDISATION-OR THE LACK OF IT?

A third issue with EV charging infrastructure in India lies in the lack of standardisation of different charger connector types. In India, currently, multiple charger types are being adopted simultaneously - CCS/ CCS Type 2, Bharat AC-001/ Bharat DC-001, CHAdeMO, GB/T, as well as in certain pockets, Tesla chargers. However, in addition to forcing prospective buyers to choose EV's in accordance with the kind of charging stations available in closest proximity to them, this also creates a discrepancy between EV manufacturers and Charging Station manufacturers/ operators. This in turn leads to inefficient, slower manufacturing, and also

acts as a bottleneck which adds to the aforementioned range and charging anxiety of EV owners.

This also creates issues on the supply side of things - As per charging guidelines issued by the Ministry of Power in 2018, most Indian manufacturers are following the DC-001 protocol, which is the most commonly found charger type in the country. However, many foreign EV manufacturers are following standards which are more popular globally, such as CCS and CHAdeMO for their vehicles. This again acts as a hindrance to their sales in India, further reducing the prospects of the growth of the EV industry.

ARE ELECTRIC VEHICLES REALLY CHEAP?

Several other issues also plague the supply side of India's EV sector. Lithium ion cells, which India does not produce, are the core element of an EV battery pack. They account for 75 percent of the cost of a battery pack, which in turn accounts anywhere between 40 and 50 percent of the cost of the EV itself. As per a Bloomberg New Energy Finance analysis, between 2020 and 2021, lithium ion battery pack prices fell from \$140 to \$132 per kilo-watt hour (KWh). However, this year has again seen a marginal increase in their prices to \$135 per KWh, owing to a mismatch in global demand and supply of lithium ion cells, and a price hike of the raw materials used in EV battery cathodes - lithium, cobalt, and nickel. Consequently, cell prices have gone up by nearly 40 percent globally, making manufacturing EV batteries and EV's themselves a lot more expensive. Increased prices result in



decreased demand in the consumer market which further constrains the adoption of EVs in India.

IS OUR WORKFORCE SKILLED ENOUGH YET?

India as a nation, for the most part, is still dependent on conventional fuels such as gasoline, diesel, and CNG for mobility. While EV sales in India have picked up, with Tata Motors, India's largest EV manufacturer, clocking record sales of over 2000 units of its Nexon and Tigor EV, a major concern is the lack of a dedicated skilled workforce of service and repair technician for EVs. Most of them are proficient only in manufacturing and servicing conventionally powered



vehicles. The costs and time associated with imparting skill training to existing as well as new workers to service EVs further adds to the delay in their adoption in India.

A lot of work is needed to be able to achieve the 30% passenger vehicle market penetration rate by 2030 target set for EVs by the Union Minister for Road Transport and Highways of India, Nitin Gadkari. A multitude of issues relating to infrastructure upgrades, skill training, supply chain logistics, as well as self dependency in component manufacturing to reduce costs need to be addressed. Without doing the same, the adoption of EVs will remain concentrated in small pockets in the country as it is now, and may remain a far-fetched dream in the larger context. **d**

	N. America	Japan	EU and the rest of markets	China	All Markets except EU
AC	 J1772 (Type 1)	 J1772 (Type 1)	 Mennekes (Type 2)	 GB/T	
DC	 CCS1	 CHAdeMO	 CCS2	 GB/T	